

ASSIGNMENT #03

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
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Due Date: Friday November 14th, my office (6:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Upload your Aspen Plus Simulation file (*.apw)
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

To perform the task of creating a Bio-Oil using the initial $50,000 \text{ kg/hr}$, we carry out a process involving pyrolysis.

As this process involves a number of units to design, they have been listed in the table below:

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Name	Unit Operator	Purpose
DRY-100	Dryer	To ensure the biomass is completely dry before entering the reactor
CYC-100 CYC-200 CYC-201A CYC-201B	Splitter (SSplit)	Divides streams based on splits specified in the outlet streams
R-200	Reactor (R-Yield)	This is the pyrolysis reactor it is a nonstoichiometric reactor based on a known yield
Mix-200	Mixer	Mixes the streams coming out of the reactor, moisture, and fluid gas streams
E-200 E-202 E-203 E-204 E-206	Heater	Increase the outlet temperature of the stream before going into the next unit operation
E-201 E-205	Heat Exchangers	Transfer of heat from one medium to another to either cool or heat the streams
R-201A R-201B	Reactor (Rstoich)	Based on known fractional conversions. With the first reactor being used for the actual reaction and the second reactor just for combustion.
ST-200	Compressor	Increases the pressure of the streams.
SEP-200	Separator (Sep2)	Separates components into two streams. Based on flows and purities.
Dec-200	Decanter	Continuously separate 2 immiscible liquids

The diagram illustrates a biomass gasification process. Biomass enters a dryer (DRY-100) where it is dried to 100. The dried biomass (DRYBIO) is then mixed with moisture (MIX-200) and heated in a heat exchanger (E-200) before entering a gasifier (201). The gasifier produces fluid gas (FLUIDGA2) and hot sand (HOTSAND1). The hot sand is cooled in a heat exchanger (E-205) and then in a separator (SEP-201) before being recycled. The fluid gas is cooled in a heat exchanger (E-202) and then in a separator (SEP-201) before being recycled. The gasifier also produces bio-oil and aqueous phases. The bio-oil is cooled in a heat exchanger (E-204) and then in a separator (SEP-201) before being recycled. The aqueous phase is cooled in a heat exchanger (E-205) and then in a separator (SEP-201) before being recycled. The gasifier also produces ash, which is cooled in a heat exchanger (E-206) and then in a separator (SEP-201) before being recycled. The gasifier also produces hot air, which is cooled in a heat exchanger (E-205) and then in a separator (SEP-201) before being recycled. The gasifier also produces flue gas, which is cooled in a heat exchanger (E-205) and then in a separator (SEP-201) before being recycled.

Legend:

- Duty (kW)
- Mass Flow (kg/h)
- △ Temperature (°C)

Explanation

Specifications PSD Entrainment Mass/Heat Transfer Atomization Drying Curve Convergence Utility Comments

Operation mode: Continuous

Dryer type: Shortcut

Operation specification

Pressure: 0 atm

Heat Duty: kW

Superheat: C

Temperature: 121.111 C

Temperature change: C

Exit stream moisture contents

Moisture specification basis: WET

Solid substream: H2O

CISOLID: 1e-10

NC: 1e-10

Valid phases: Vapor-Liquid

Figure 2: Dryer Unit Specifications

This enables the SSplit unit directly linked to the output stream of the Dryer unit to be able to discern between the moisture component and dry biomass coming from the feed, so that the only components going into the R-Yield reactor are the dry biomass.

The specification for the SSplit is shown below:

Specifications Flash Options Key Components Comments

Specification for each substream

Stream names: MOISTURE

Substream Name	Specification	Basis	Value	Units	Key Comp No
MIXED	Split fraction		1		
CISOLID	Split fraction		0		
NC	Split fraction		0		

Figure 3: Outlet MOISTURE stream specification of SSplit

Thus, the only component in the MOISTURE stream is the one specified, H_2O . Now our R-Yield can process the dry components with ease, given that the yield specifications are in the problem statement. Along with this, there is an inert component: Sand, which enters the R-Yield reactor in order to deliver the energy required to carry out the pyrolysis reactions.

The R-Yield reactor specifications follow as listed:

Specifications Streams Yield Flash Options PSD Comp. Attr. Comp. Mapping

Operating conditions

Flash Type **Temperature** **Pressure**

Temperature **480** **C**

Temperature change **C**

Pressure **-0.303975** **bar**

Duty **cal/sec**

Vapor fraction

Valid phases

Vapor-Liquid

Figure 4: R-Yield reactor Specifications

Specifications Streams Yield Flash Options PSD Comp. Attr.

Yield specification

Yield options **Component yields**

Component yields

Component	Basis	Basis Yield
H2	Mass	0.1
CO	Mass	5.82
CO2	Mass	6.03
CH4	Mass	0.53
C2H4	Mass	0.42
H2O (MIXED)	Mass	14.8
CHAR	Mass	9.7
AACID	Mass	11.1
ACETONE	Mass	12.7
MCRESOL	Mass	4
GUAIACOL	Mass	27.5
LVG	Mass	4.4
FURFURAL	Mass	2.9

Inert Components

SAND

Figure 5: R-Yield, Yield specifications

☒ Specifications
 ☐ Streams
 ☒ Yield
 ☐ Flash Options
 ☐ PSD
 ☒ Comp. Attr.
 ☐ Com

Substream ID ☒ NC

Component attributes

Component ID ☒ CHAR

Attribute ID ☒ PROXANAL

Element	Value
MOISTURE	
FC	84.745
VM	14.955
ASH	0.3

☒ Specifications
 ☐ Streams
 ☒ Yield
 ☐ Flash Options
 ☐ PSD
 ☒ Comp. Attr.
 ☐ Co

Substream ID ☒ NC

Component attributes

Component ID ☒ CHAR

Attribute ID ☒ ULTANAL

Element	Value
ASH	0.3
CARBON	49.66
HYDROGEN	6.31
NITROGEN	0.1
CHLORINE	0
SULFUR	0.08
OXYGEN	43.55

☒ Specifications
 ☐ Streams
 ☒ Yield
 ☐ Flash Options
 ☐ PSD
 ☒ Comp. Attr.
 ☐ Co

Substream ID ☒ NC

Component attributes

Component ID ☒ CHAR

Attribute ID ☒ SULFANAL

Element	Value
PYRITIC	0
SULFATE	0.04
ORGANIC	0.04

Figure 6: Comp. Attributes for Char

Once we get our products from the R-Yield, they are once mixed again with the once-removed Moisture component in the mixer (MIX-200). Here we also add a flue gas stream that essentially help to carry the solids in the streams and units.

The now mixed streams go towards a “fake” heater unit holding the specifications:

Flash specifications

Flash Type: Temperature, Pressure

Temperature: 480 C

Temperature change: C

Degrees of superheating: C

Degrees of subcooling: C

Pressure: 0 bar

Duty: cal/sec

Vapor fraction:

Pressure drop correlation parameter:

☐ Always calculate pressure drop correlation parameter

Valid phases: Vapor-Liquid

Figure 7: Specifications of "fake" heater unit

This stream is then sent into the SSplit unit (CYC-200) where the stream is once again split into biomass and the remaining components.

Specification for each substream

Stream names: 203A

Substream Name	Specification	Basis	Value	Units	Key Comp No
MIXED	Split fraction		1		
CISOLID	Split fraction		0		
NC	Split fraction		0		

Figure 8: CYC-200 specifications.

The biomass (203B) is sent to the pyrolysis reactor where the breakdown of char into ash will be carried out. When specifying the reactor we initially put a coefficient guess of 1 for each component, however this is overwritten by the calculator (C-RXN) that takes the molar mass of the participating units and manages to find the number of moles required for the reaction. Stream 203B goes into our char combustor reactor.

✕ Edit Stoichiometry

Reaction No. 1

Reactants	
Component	Coefficient
CHAR	-1

Products	
Component	Coefficient
O2	1
N2	1
C (CISOLID)	1
H2	1
SULFUR	1
ASH	1

Products generation

☐ Molar extent

☒ Fractional conversion of component

Close

Figure 9: Reaction taking place in R-201A

Variable	Information flow	Definition
MO2	Import variable	Unary-Param Variable=MW ID1=O2 ID2=1
MN2	Import variable	Unary-Param Variable=MW ID1=N2 ID2=1
MC	Import variable	Unary-Param Variable=MW ID1=C ID2=1
MH2	Import variable	Unary-Param Variable=MW ID1=H2 ID2=1
MS	Import variable	Unary-Param Variable=MW ID1=SULFUR ID2=1
NO2	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=O2 ID3=MIXED
NN2	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=N2 ID3=MIXED
NC	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=C ID3=CISOLID
NASH	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=ASH ID3=NC
NH2	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=H2 ID3=MIXED
NS	Export variable	Block-Var Block=R-201A Variable=COEF Sentence=STOIC ID1=1 ID2=SULFUR ID3=MIXED
MOIST	Import variable	Compattr-Var Stream=203B Substream=NC Component=CHAR Attribute=PROXANAL Ele...
ULTIMATE	Import variable	Compattr-Vec Stream=203B Substream=NC Component=CHAR Attribute=ULTANAL

Figure 10: List of Variables for calculation

```

W0= MOIST/100
DRY=1-W0
NO2=DRY*ULTIMATE (7) /MO2/100.0
NN2=DRY*ULTIMATE (4) /MN2/100.0
NASH=DRY*ULTIMATE (1) /100.0
NC=DRY*ULTIMATE (2) /MC/100.0
NH2=DRY*ULTIMATE (3) /MH2/100.0
NS=DRY*ULTIMATE (6) /MS/100.0

```

Figure 11: Fortran code for calculations

This allows for the combustion reaction in the next R-Stoic reactor (R-201B) with the addition of a hot air inlet that is heated by E-205. Doing so allows us to separate the product stream into three components: Fluegas – which can be used to heat up the air inlet, Fine Ash – one of the products and finally, the hot sand we had used in the R-Yield reactor beforehand.

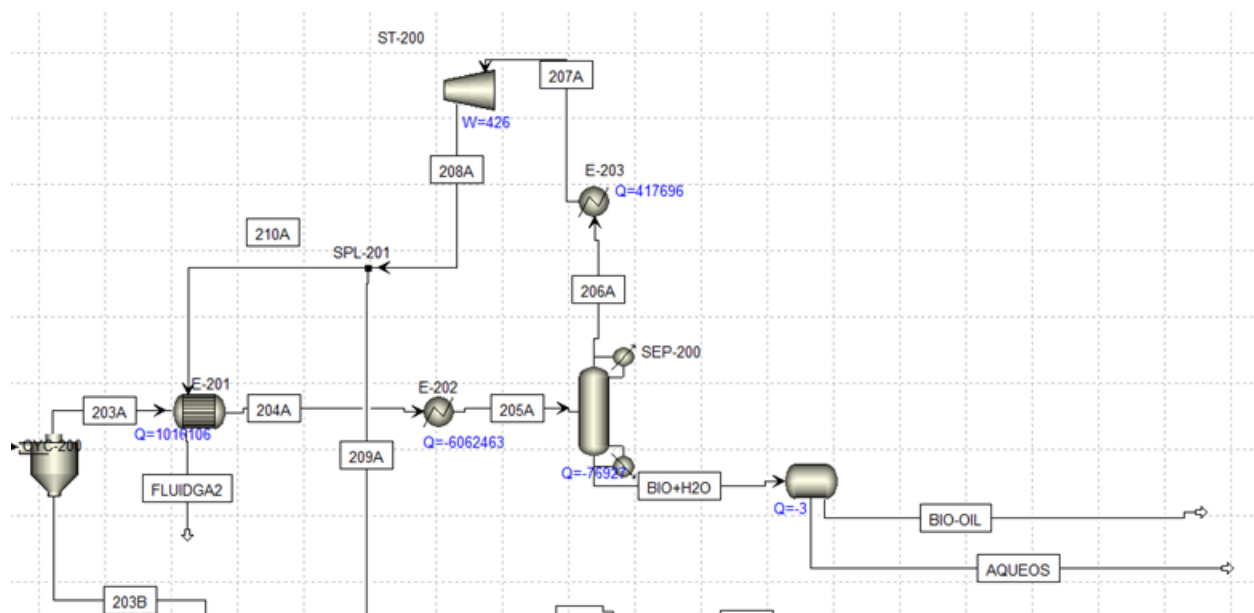


Figure 8: Top section of splitter

Looking at the top section of the splitter, we have the following PFD. After separating the mixed sub stream from the CISOLID and NC.

Model fidelity	Hot fluid	Shortcut flow direction
☒ Shortcut	☐ Shell	☒ Countercurrent
☐ Detailed	☐ Tube	☐ Cocurrent
☐ Shell & Tube		☐ Multipass, calculate number of shells
☐ Kettle Reboiler		☐ Multipass, shells in series 1
☐ Thermosyphon		
☐ Air Cooled		
☐ Plate		

Calculation mode: **Design**

Exchanger specification

Specification: **Cold stream outlet temperature**

Value: **375 C**

Exchanger area: sqm

Constant UA: cal/sec-K

Minimum temperature approach: **2 C**

Figure 9: E-201 specifications

E-201 is a heat exchanger that is capturing the heat from the hot stream of 203A in order to heat the fluidized gas before it enters the pyrolysis riser. This prevents cold gas from cooling the hot

sand and dropping the pyrolysis temperature. Improving energy efficiency and keeping the reactor operating at the correct temperature. Now in our PDF we do not have FLUIDGA2 going straight into the pyrolysis reactor as when we attempted this ASPEN would crash every time. Which is why we have a separate FLUIDGAS stream going into MIX-100

The screenshot shows the 'E-202 (Heater)' specifications window. The 'Flash specifications' section is active, showing the following settings:

Parameter	Value	Unit
Flash Type	Temperature	
Flash Type	Pressure	
Temperature	45	C
Temperature change		C
Degrees of superheating		C
Degrees of subcooling		C
Pressure	1.72253	bar
Duty	0	cal/sec
Vapor fraction	1	
Pressure drop correlation parameter		

Below the specifications, there is a checkbox labeled 'Always calculate pressure drop correlation parameter' which is checked. At the bottom, the 'Valid phases' section shows 'Vapor-Liquid' selected.

Figure 10: E-202 Specifications

Stream 204A is still hot coming out of the heat exchanger. With a Need to cool it down even further before entering the separator. Using a condenser with the design specs of 45°C and 1.7 bar as these are the conditions that the stream needs to be as given by the problem statement.

Main Flowsheet x SEP-200 (Sep2) x +

Specifications Feed Flash Outlet Flash Utility Comments

Substream MIXED Outlet stream 206A

Stream spec Split fraction

Component ID	1st Spec
H2	Split fraction 1
N2	Split fraction 1
CO	Split fraction 1
O2	Split fraction 1
CH4	Split fraction 1
C2H4	Split fraction 1
CO2	Split fraction 1
SO2	Split fraction 1
NO2	Split fraction 1
ACETONE	Split fraction 0.96
2MEFURAN	Split fraction 0
BENZENE	Split fraction 0
H2O	Split fraction 0.2
TOLUENE	Split fraction 0
AACID	Split fraction 0
CYPENTON	Split fraction 0
PXYLENE	Split fraction 0
FURFURAL	Split fraction 0
BZFURAN	Split fraction 0
INDANE	Split fraction 0
INDENE	Split fraction 0
5MEFURAN	Split fraction 0
MCRESOL	Split fraction 0
GUAIACOL	Split fraction 0
MNAPTHEN	Split fraction 0
NAPHTHOL	Split fraction 0
CONALDE	Split fraction 0
LVG	Split fraction 0

Specifications Feed Flash Outlet Flash Utility Comments

Inlet flash specifications

Pressure 1.72253 bar

Valid phases Vapor-Liquid

Flash options

Maximum iterations 30

Error tolerance 0.0001

Figure 11: Specifications of SEP-200

These are the specifications that we gave the SEP-200. In this separator we assigned split fractions to control exactly which components leave with each outlet stream. All permanent gases such as H₂, CO, CO₂, CH₄, N₂ are given a split fraction of 1. This means that they fully exit with the vapor outlet because they can't condense. The heavy bio-oul compounds were all

given a split fraction of 0. This makes sure they stay in the liquid outlet. Water is given an intermediate split as well as acetone to show partial vaporization.

Coming out the bottom we have BIO+H₂O which needs to get separated. Using a decanter with the following specifications.

The top screenshot shows the 'Decanter specifications' tab. Under 'Decanter specifications', the 'Pressure' is set to 0 bar, and 'Duty' is set to 0 cal/sec. Under 'Key components to identify 2nd liquid phase', 'H2O' is selected as the key component from the 'Available components' list.

The bottom screenshot shows the 'Efficiency' tab. It displays a table for 'Separation efficiency for each component'.

Component	Efficiency
H2	
N2	
CO	
O2	
CH4	
C2H4	
CO2	
SO2	
NO2	
ACETONE	
2MEFURAN	
BENZENE	
H2O	100

Figure 12: Specifications for Decanter

A decanter is used because the condensed liquid leaving separator 2 naturally splits into two immiscible liquid phases. Organic bio-oil and aqueous. Specifying H₂O as the key component to identify the second liquid phase because water forms the aqueous layer from the organic bio-oil. The decanter's separation efficiency for water is set to 100%. This makes sure that all the condensed water goes to the aqueous outlet and does not remain mixed with the bio-oil phase.

Coming out of the distillate there is E-203 and ST-200 which are a heater and compressor respectively. These are used to increase the pressure and temperature of the stream. Which proceeds to go into SPL-201. Stream 210A goes through the E-201 and gives us our FLUIDGA2. Stream 209A goes into R-201B, our second stoich reactor.

After looking over the entire process we have 2 more design specs set up

The screenshot displays the Aspen Plus Design Specification Editor interface. The 'Define' tab is active, showing a table of sampled variables. The variable 'ADIABATI' is defined with the expression 'Block-Var Block=R-200 Variable=QCALC Sentence=PARAM Units=kW'. Below the table, the 'Edit selected variable' section shows the variable 'ADIABATI' selected. The 'Reference' section shows the configuration: Type: Block-Var, Block: R-200, Variable: QCALC, Sentence: PARAM, Units: kW. The 'Design specification expressions' section shows the Spec: ADIABATI, Target: 0, and Tolerance: 0.001. The 'Manipulated variable' section shows the configuration: Type: Block-Var, Block: E-206, Variable: TEMP, Sentence: PARAM, Units: C. The 'Manipulated variable limits' section shows the Lower limit: 200 and Upper limit: 800.

Variable	Definition
ADIABATI	Block-Var Block=R-200 Variable=QCALC Sentence=PARAM Units=kW

Buttons: New, Delete, Copy, Paste, Move Up, Move Down, View Variables

Edit selected variable

Variable: ADIABATI

Category: All, Blocks, Streams

Reference

Type: Block-Var

Block: R-200

Variable: QCALC

Sentence: PARAM

Units: kW

Design specification expressions

Spec: ADIABATI

Target: 0

Tolerance: 0.001

Manipulated variable

Type: Block-Var

Block: E-206

Variable: TEMP

Sentence: PARAM

Units: C

Manipulated variable limits

Lower: 200

Upper: 800

Step size:

Maximum step size:

Report labels:

Figure 12: Design Spec for incoming sand

This calculator is used to change the temperature of the sand, exiting the heat exchanger.

	Variable	Information flow	Definition
▶	TSAND1	Import variable	Stream-Var Stream=HOTSAND2 Substream=CISOLID Variable=TEMP Units=C
▶	TSAND2	Export variable	Stream-Var Stream=HOTSAND1 Substream=NC Variable=TEMP Units=C
*			

Enter executable Fortran statements

```
TSAND2=TSAND1
```

Figure 13: Sand Calculator

This calculator is used to calculate the temperature of the sand that is exiting E-206 and entering R-200. They are set equal since the HOTSAND2 is supposed to enter R-200 but again it makes our aspen crash, so it is separated.

References:

CHE 433: Process Simulation with Aspen Plus - Chapter 5 Physical Property Methods

CHE 433: Process Simulation with Aspen Plus - Chapter 6 Chemical Reactors and Non Conventional Solids

CHE 433: Process Simulation with Aspen Plus - Chapter 7 Distillation Unit Operations

ChatGPT

